

Geometric Welfare Comparison of Assignment Mechanisms in Finite Markets

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Abstract

This thesis develops a framework for comparing Probabilistic Serial and Random Serial Dictatorship. Abel summation turns each agent's expected-utility gap into $d^\top \Delta_i$, where d is a vector of adjacent-rank utility decrements and Δ_i is the cumulative rank-lottery gap between PS and RSD. The main object is the utility-sensitive set: the region in which admissible utilities disagree about whether Probabilistic Serial or Random Serial Dictatorship delivers higher welfare. Rather than treating this region as an unresolved residual, the paper characterises it geometrically using lower and upper envelope functionals over admissible decrement families. For structured utility restrictions, including front-weighted, tail-weighted, and tiered decrement cones, robust dominance and utility sensitivity reduce to finite coordinate, prefix, suffix, and block-sum certificate checks.

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1 Introduction

Random assignment mechanisms allocate indivisible objects when agents have ordinal preferences and monetary transfers are unavailable or inappropriate. This setting appears in school choice, course allocation, housing assignment, and related matching problems. In such environments, the designer must compare mechanisms that embody different normative priorities. Two central benchmarks are Probabilistic Serial (PS) and Random Serial Dictatorship (RSD). Probabilistic Serial is attractive for its ordinal efficiency and fairness properties [3], while Random Serial Dictatorship is attractive for its simplicity, strategy-proofness, and its close connection to random priority [1]. Understanding how to compare these mechanisms is therefore a central question in random assignment.

The difficulty is that welfare comparison in this setting is not determined by ordinal rankings alone. The two mechanisms generally produce different lotteries over ranks. One mechanism may raise the probability of receiving a top-ranked object, while the other may reduce the probability of receiving a very poor outcome. Which improvement matters more depends on the cardinal utility attached to ranks. Ordinal preferences tell us that better ranks are preferred to worse ones, but they do not tell us how steeply utility falls across ranks. As a result, the welfare ranking of PS and RSD is often ambiguous unless one imposes additional structure on admissible utility representations.

This paper studies that ambiguity directly. Rather than selecting a single cardinalisation of ordinal preferences, it asks when the welfare comparison between PS and RSD is robust over a class of admissible rank utilities, and when it is genuinely utility-sensitive. In other words, the paper asks when every utility in a given admissible family ranks one mechanism above the other, and when different admissible utilities lead to different welfare conclusions.

The starting point is a finite-dimensional representation of the comparison. For each agent, the difference between the two mechanisms can be summarised by a cumulative rank-lottery gap vector, which records at each rank cutoff how much more probability PS assigns than RSD to receiving an object at least that good. Abel summation then shows that the expected-

utility difference between the two mechanisms can be written as an inner product between this cumulative gap vector and the vector of adjacent-rank utility decrements. This separates the problem into two parts: the mechanism and preference profile determine the cumulative gap, while a policymaker or analyst specifies which decrement vectors are admissible.

The paper’s first contribution is to use this representation to develop a finite restricted stochastic dominance framework. For an admissible family D of decrement vectors, the paper defines lower and upper envelope functionals that determine whether the welfare sign is unanimous over D , unanimously reversed over D , or mixed within D . This yields a robust region favoring PS, a robust region favoring RSD, and a utility-sensitive region in which admissible utilities disagree. A central theme of the paper is that this utility-sensitive region is not merely a leftover category. It is a structured geometric object that captures the exact form of finite-market welfare indeterminacy.

The second contribution is tractability. The paper studies cones of admissible decrement vectors that correspond to economically interpretable restrictions on utility intensity. The full cone reproduces first-order stochastic dominance. Front-weighted cones capture settings in which improvements near the top of the ranking matter most. Tail-weighted cones capture settings in which avoiding poor outcomes matters most. Tiered cones capture environments in which outcomes are naturally grouped into broad categories such as excellent, acceptable, and unacceptable. For these classes, the paper derives exact finite criteria. Robust dominance reduces to coordinatewise, prefix-sum, suffix-sum, or block-sum checks, and the same finite certificates characterise the utility-sensitive region. If a normalisation of utility functions assumption is held, this could be extrapolated to polytopes in general not necessarily cones.

This restrictions are inspired by economically meaningful intuition rather than abstract mathematical intuition. For example, in a school choice setting agents may care proportionally more about the top ranked schools. On the other hand, people may care a lot about not getting unacceptable housing and may not care as much between the top ranked choices. In such settings, we should look at which mechanism dominates over an admissible class of

utilities tied to the priorities and nature of the institution, rather than more mathematically abstract view on whether one mechanism dominates under every decreasing utility function (First Order Stochastic Dominance).

The simulations in the paper serve to illustrate how the theory could be implemented on a synthetic profile. It is limited in use since it doesn't constitute real market validation, but it demonstrates the practical content of the theory in a more intuitive light.

2 Related Literature

This thesis lies at the intersection of 3 literatures. The random assignment literature, where the mechanisms originate. Robustness criterion closely aligns with the stochastic dominance literature. The literature on convex geometry and welfare aligns with the linear welfare functionals, half spaces and finite certificates in this thesis.. This thesis combines these groups into a finite restricted dominance framework for comparing PS and RSD.

2.1 Random Assignment

The random assignment problem refers to how to allocate indivisible goods when agents report ordinal preferences without monetary transfers. It originates in the theoretical and statistical computer science literature but has certainly spilled over into economics as it is analogous to the basic economic problem. Hylland and Zeckhauser [10] provide a cardinal benchmark of ex-ante Pareto efficiency by demonstrating how to optimally allocate indivisible goods based on exact cardinal utilities. This paper does not use cardinalisation as an explicit mechanism input, rather uses structured class of utility decrements to resolve welfare incomparability inherent to purely ordinal systems.

Bogomolnaia and Moulin [3] provides the second assignment mechanism PS and also formalise ordinal efficiency in the random assignment problem context. They define ordinal efficiency by a random assignment is ordinally efficient if no agent can be made better off under

first-order stochastic dominance without harming another.. Bogomolnaia and Moulin show that the rank lotteries induced by PS and RSD are frequently incomparable under first-order stochastic dominance. This thesis explores the expected utility comparison depends on the cardinal utility used to represent ordinal ranks, motivating the restricted utility framework.

Subsequent work develops the domain, axioms, and asymptotics of these mechanisms. Katta and Sethuraman [11] extend the PS idea to weak preferences, showing that the ordinal-efficiency logic can be adapted beyond strict rankings. Hashimoto, Hirata, Kesten, Kurino, and Unver [9] give axiomatic approaches to PS, clarifying the normative principles behind the mechanism. Che and Kojima [4] show that PS and random priority become asymptotically equivalent in large assignment markets with many copies, while Kojima and Manea [12] study the incentives of PS in large economies. Manea [16] shows that the ordinal inefficiency of RSD can remain important in large settings. These papers explain why large markets can soften the finite PS/RSD contrast, but they do not settle the finite-market comparison studied here.

2.2 Stochastic Dominance

A second relevant literature connects ordinal efficiency to incomplete cardinal information.

McLennan [18] ordinal efficiency welfare theorem serves as an important reference point for this thesis because it uses convex geometry to connect ordinal assignment properties with cardinal utility representations. McLennan proves an ordinal-efficiency welfare theorem using the polyhedral separating hyperplane theorem McLennan establishes an existential result across at least one supporting, compatible cardinalisation. This result provides crucial theoretical comfort, ensuring ordinal efficiency is anchored to traditional cardinal welfare in at least a best-case scenario. This thesis builds indirectly on this, investigating compatible classes of utility functions using robust systemic properties versus a fragile utility-sensitive artifact.

Dogan, Dogan, and Yildiz [5] show that PS weakly dominates RSD in aggregate social

welfare. This weak dominance leaves open the agent-level question of when and for whom PS robustly delivers higher expected utility across institutionally defensible utility classes, which is the question this thesis addresses.

Aziz, Brandl, and Brandt[2] once again proves the existence of a general principle: a lottery is robustly welfare superior across a set of plausible utility functions if and only if it is affine welfare maximising for some vector in that set. This thesis aims to operationalise this principle by restricting utility classes into institutionally motivated decrement cones to derive finite certificate conditions

Stochastic dominance is the natural robustness benchmark. Hanoch and Levy [7] and Levy [14] develop the expected-utility interpretation of stochastic dominance: distributional inequalities matter because they correspond to unanimous rankings over a class of utility functions. In rank-lottery form, first-order stochastic dominance requires that one mechanism give at least as much probability of a top- k assignment for every rank cutoff k . This is exactly the full-class benchmark in the thesis: robustness over all decreasing rank utilities.

Full first-order stochastic dominance is powerful but demanding. If the cumulative rank-lottery gaps change sign across cutoffs, the full class of decreasing utilities does not rank the two lotteries. The literature on restricted and almost stochastic dominance explains why analysts may still want meaningful comparisons after full dominance fails. Leshno and Levy [13] argue that classical dominance can be too strict because it lets extreme utility functions veto comparisons preferred by most decision makers. Fishburn and Lavalley [6] study stochastic dominance on finite grids, which is especially relevant because assignment ranks are discrete. This thesis keeps the unanimity logic of stochastic dominance but looks at different admissible classes that are more realistic scenarios.

2.3 Convex Geometry

The final ingredient is convex geometry. Expected utility is an inner product between a lottery and a utility vector. In this thesis, Abel summation rewrites that inner product as $d^\top \Delta_i$,

separating the mechanism-generated cumulative rank-lottery gap Δ_i from the admissible utility-decrement vector d . This is why half-spaces and cones appear naturally. For a fixed utility class D , robust dominance is the requirement that $d^\top \Delta_i$ have the same sign for every $d \in D$. When D is a polyhedral cone, this becomes a finite certificate problem. The majorisation literature, including Hardy, Littlewood, and Polya [8] and Marshall, Olkin, and Arnold [17], provides a useful analogy through cumulative-sum inequalities, but the thesis does not claim that its rank-lottery comparisons are simply majorisation results. The shared structure is the use of cumulative inequalities and linear certificates.

2.4 Positioning and Contribution

The closest literatures provide the mechanisms, the robustness principle, and the convex tools, but not the specific finite-market diagnostic developed here. This thesis combines the 3 literatures for a specific comparison: for a realised cumulative rank lottery gap, deriving finite tests for robustness and utility sensitivity in identified admissible decrement classes.

The main contribution is therefore operational and geometric. Full first-order stochastic dominance appears as the benchmark case in which all nonnegative decrement vectors are admissible. Front-weighted, tail-weighted, and tiered utility classes are then treated as economically interpretable restrictions on adjacent-rank losses. For these structured classes, robust comparison reduces to coordinate, prefix, suffix, block-prefix, and block-suffix certificate inequalities. This allows a policymaker to see exactly when a welfare ranking is invariant over an admissible utility class and when the comparison genuinely depends on cardinalisation

This framing also clarifies the thesis’s limits. It does not claim that PS universally welfare-dominates RSD. It does not claim that RSD’s ex post efficiency is a cardinal welfare guarantee. It does not claim novelty for expected utility, Abel summation, stochastic dominance, separating hyperplanes, dual cones, normalised polytopes, or the mechanisms themselves. Its contribution is to assemble these tools into a finite restricted-dominance framework for comparing two canonical assignment mechanisms, with explicit certificates for

economically meaningful utility domains.

3 Model and Welfare Representation

There are n agents and n objects. For mechanism $M \in \{\text{PS}, \text{RSD}\}$, let $p_i^M(k)$ be the probability that agent i receives her rank- k object. Define the cumulative rank lottery

$$F_i^M(k) = \sum_{r=1}^k p_i^M(r), \quad k = 1, \dots, n.$$

The cumulative gap between PS and RSD is

$$\Delta_i(k) = F_i^{\text{PS}}(k) - F_i^{\text{RSD}}(k), \quad k = 1, \dots, n-1.$$

The final cutoff is omitted because both cumulative probabilities equal one.

Let w be a weakly decreasing rank utility,

$$w(1) \geq w(2) \geq \dots \geq w(n),$$

and define the adjacent-rank decrement vector

$$d_w(k) = w(k) - w(k+1), \quad k = 1, \dots, n-1.$$

Proposition 1 (Abel representation). *For every agent i and every decreasing rank utility w ,*

$$\text{EU}_i^{\text{PS}}(w) - \text{EU}_i^{\text{RSD}}(w) = d_w^\top \Delta_i.$$

This proposition is the bridge from rank lotteries to linear welfare comparison. It shows that every admissible cardinal comparison reduces to a linear functional evaluated on the cumulative-gap vector.

Proof sketch. Write the rank-probability difference as $a_k = p_i^{\text{PS}}(k) - p_i^{\text{RSD}}(k)$, take partial sums $A_k = \sum_{r \leq k} a_r = \Delta_i(k)$, and apply summation by parts.

4 Roadmap of Theory

Result	Mathematical content	Role in paper
Prop. 1: Abel representation	$\text{EU}_i^{\text{PS}}(w) - \text{EU}_i^{\text{RSD}}(w) = d_w^\top \Delta_i$.	Converts welfare comparison into a linear comparison between utility decrements and cumulative rank-lottery gaps.
Thm. 1: restricted dominance	Robust-PS iff $\phi_D(\Delta_i) \geq 0$; robust-RSD iff $\psi_D(\Delta_i) \leq 0$.	Defines robust regions, equality, and utility sensitivity for any admissible decrement family.
Cor. 1: cone envelopes	If $D \subseteq K$, robustness over K implies robustness over D .	Explains why cones can act as conservative admissible envelopes.
Prop. 2: enlargement	If $D_0 \subseteq D_1$, robust regions shrink and sensitivity weakly expands.	Shows how broadening utility classes makes robust conclusions harder.
Prop. 3: dual cone	For closed convex K , robust-PS iff $\Delta_i \in K^*$.	Turns robust welfare comparison into cone membership in Δ -space.
Prop. 4: finite generators	If $K = \text{cone}(g^1, \dots, g^m)$, check only $(g^j)^\top \Delta_i$.	Turns infinite utility classes into finite certificate checks.
Prop. 5: geometry of $\mathcal{S}_D$	$\Delta \in \mathcal{S}_D$ iff some admissible decrements give opposite signs.	Shows utility sensitivity is sign crossing, not a residual category.
Lemma 1: generators	Structured cones are generated by coordinate, prefix, suffix, and block-step vectors.	Provides the algebraic basis for the certificate criteria.
Thm. 2: structured criteria	Robust-PS reduces to coordinate, prefix, suffix, or block-sum nonnegativity.	Main tractability result for economically interpretable utility restrictions.
Cor. 2: sign crossing	For structured cones, $\Delta_i \in \mathcal{S}_D$ iff certificates have mixed signs.	Makes utility sensitivity directly computable in simulations.
Cor. 3: full cone/FOSD	Robust dominance over K^{all} iff $\Delta_i(k) \geq 0$ for all k .	Embeds first-order stochastic dominance as the full-utility benchmark.
Cor. 4: non-nesting	K^{front} and K^{tail} are not nested.	Clarifies that front and tail restrictions encode different priorities.
Cor. 5: aggregate dominance	Apply the same cone logic to $\bar{\Delta} = \frac{1}{n} \sum_i \Delta_i$.	Extends the framework from agent-level to average welfare comparisons.

Table 1: This table explains the content of the theory in its simplest form

5 Admissible Families, Cone Envelopes, and Utility Sensitivity

Let $D \subseteq \mathbb{R}_+^{n-1}$ be a nonempty admissible family of decrement vectors. Agent i is robust-PS over D if

$$d^\top \Delta_i \geq 0 \quad \text{for every } d \in D,$$

and robust-RSD over D if

$$d^\top \Delta_i \leq 0 \quad \text{for every } d \in D.$$

Theorem 1 (Restricted dominance over admissible families). *For any nonempty $D \subseteq \mathbb{R}_+^{n-1}$, define*

$$\phi_D(\Delta) = \inf_{d \in D} d^\top \Delta, \quad \psi_D(\Delta) = \sup_{d \in D} d^\top \Delta.$$

Then agent i is robust-PS over D if and only if

$$\phi_D(\Delta_i) \geq 0,$$

and robust-RSD over D if and only if

$$\psi_D(\Delta_i) \leq 0.$$

If

$$\phi_D(\Delta_i) < 0 < \psi_D(\Delta_i),$$

then the comparison is utility-sensitive over D . If

$$\phi_D(\Delta_i) = \psi_D(\Delta_i) = 0,$$

then $d^\top \Delta_i = 0$ for every $d \in D$.

This theorem gives the exact D -level notion of restricted dominance. The lower and upper envelopes summarise whether the entire admissible family signs the welfare comparison in one direction, the other direction, or both.

Proof sketch. The envelopes simply repackage the universal sign conditions over D . Utility sensitivity is exactly the case where some admissible decrement vectors give a negative welfare gap and others give a positive one.

Corollary 1 (Cone envelopes). *Let $D \subseteq K \subseteq \mathbb{R}_+^{n-1}$. If agent i is robust-PS over K , then agent i is robust-PS over D . Likewise, if agent i is robust-RSD over K , then agent i is robust-RSD over D .*

This corollary explains why cones are useful admissible envelopes. A positive result on a larger cone automatically yields the same result on any smaller normalised or parametric family contained inside it.

This thesis will use:

$$S_D = \{\Delta \in \mathbb{R}^{n-1} : \phi_D(\Delta) < 0 < \psi_D(\Delta)\}$$

for the utility-sensitive region associated with the admissible family D .

Proposition 2 (Envelope conservatism under admissible-family enlargement). *Let $D_0 \subseteq D_1 \subseteq \mathbb{R}_+^{n-1}$ be nonempty admissible families. Then for every $\Delta \in \mathbb{R}^{n-1}$,*

$$\phi_{D_1}(\Delta) \leq \phi_{D_0}(\Delta), \quad \psi_{D_1}(\Delta) \geq \psi_{D_0}(\Delta).$$

Consequently,

$$\{\Delta : \phi_{D_1}(\Delta) \geq 0\} \subseteq \{\Delta : \phi_{D_0}(\Delta) \geq 0\},$$

$$\{\Delta : \psi_{D_1}(\Delta) \leq 0\} \subseteq \{\Delta : \psi_{D_0}(\Delta) \leq 0\},$$

and

$$\{\Delta : \phi_{D_0}(\Delta) < 0 < \psi_{D_0}(\Delta)\} \subseteq \{\Delta : \phi_{D_1}(\Delta) < 0 < \psi_{D_1}(\Delta)\}.$$

This proposition formalises the conservative role of outer envelopes: enlarging the admissible family weakly shrinks robust regions and weakly expands structured disagreement.

Proposition 3 (Dual-cone characterisation). *Let $K \subseteq \mathbb{R}_+^{n-1}$ be a closed convex cone, and define*

$$K^* = \{z \in \mathbb{R}^{n-1} : d^\top z \geq 0 \text{ for all } d \in K\}.$$

Then agent i is robust-PS over K if and only if

$$\Delta_i \in K^*.$$

Agent i is robust-RSD over K if and only if

$$-\Delta_i \in K^*.$$

This is the main cone-level bridge. Once admissible families are widened to closed convex cones, robust welfare comparison becomes a dual-cone membership question in Δ -space.

Proposition 4 (Finite-generator reduction). *Suppose*

$$K = \text{cone}(g^1, \dots, g^m)$$

for generators $g^1, \dots, g^m \in \mathbb{R}_+^{n-1}$. Then agent i is robust-PS over K if and only if

$$(g^j)^\top \Delta_i \geq 0 \quad \text{for every } j = 1, \dots, m.$$

Agent i is robust-RSD over K if and only if

$$(g^j)^\top \Delta_i \leq 0 \quad \text{for every } j = 1, \dots, m.$$

This reduction is what turns an infinite admissible cone into a finite checklist. For the structured cones later in the paper, the theorem is made fully explicit by writing down the relevant generators.

Proof sketch. If each generator has the right sign, then every nonnegative combination of them has the same sign. Conversely, each generator already belongs to the cone, so violating one certificate violates robustness.

Proposition 5 (Geometry of utility-sensitive comparisons). *For any nonempty admissible family $D \subseteq \mathbb{R}_+^{n-1}$,*

$$\Delta \in \mathcal{S}_D$$

if and only if there exist $d, d' \in D$ such that

$$d^\top \Delta < 0 < (d')^\top \Delta.$$

Equivalently, defining

$$U_d^+ = \{\Delta : d^\top \Delta \geq 0\}, \quad U_d^- = \{\Delta : d^\top \Delta \leq 0\},$$

the robust-PS region is $\bigcap_{d \in D} U_d^+$, the robust-RSD region is $\bigcap_{d \in D} U_d^-$, and

$$\mathcal{S}_D = \left(\bigcup_{d \in D} \{\Delta : d^\top \Delta < 0\} \right) \cap \left(\bigcup_{d \in D} \{\Delta : d^\top \Delta > 0\} \right).$$

This proposition identifies the utility-sensitive set as a genuine geometric object rather than a leftover category. It is the region where admissible linear welfare functionals cross the same Δ -vector from both sides.

6 Structured Decrement Cones

The full positive cone is

$$K^{\text{all}} = \mathbb{R}_+^{n-1}.$$

The front-weighted cone is

$$K^{\text{front}} = \{d \in \mathbb{R}_+^{n-1} : d(1) \geq d(2) \geq \cdots \geq d(n-1)\}.$$

The tail-weighted cone is

$$K^{\text{tail}} = \{d \in \mathbb{R}_+^{n-1} : d(1) \leq d(2) \leq \cdots \leq d(n-1)\}.$$

For tiered cones, fix an ordered partition $B_1, \dots, B_J$ of $\{1, \dots, n-1\}$ and define block sums

$$S_j(\Delta_i) = \sum_{k \in B_j} \Delta_i(k).$$

Then

$$K^{B\text{-front}} = \{d \in \mathbb{R}_+^{n-1} : d(k) = c_j \text{ for } k \in B_j, \ c_1 \geq c_2 \geq \cdots \geq c_J \geq 0\},$$

and

$$K^{B\text{-tail}} = \{d \in \mathbb{R}_+^{n-1} : d(k) = c_j \text{ for } k \in B_j, \ 0 \leq c_1 \leq c_2 \leq \cdots \leq c_J\}.$$

Lemma 1 (Generator representations). *Let e^m denote the m -th coordinate vector in $\mathbb{R}^{n-1}$.*

Define

$$q^m(k) = \mathbf{1}\{k \leq m\}, \quad r^m(k) = \mathbf{1}\{k \geq m\}.$$

For tiered cones, define

$$Q^m(k) = \mathbf{1}\{k \in B_1 \cup \cdots \cup B_m\}, \quad R^m(k) = \mathbf{1}\{k \in B_m \cup \cdots \cup B_J\}.$$

Then

$$K^{\text{all}} = \text{cone}\{e^1, \dots, e^{n-1}\},$$

$$K^{\text{front}} = \text{cone}\{q^1, \dots, q^{n-1}\}, \quad K^{\text{tail}} = \text{cone}\{r^1, \dots, r^{n-1}\},$$

and

$$K^{B\text{-front}} = \text{cone}\{Q^1, \dots, Q^J\}, \quad K^{B\text{-tail}} = \text{cone}\{R^1, \dots, R^J\}.$$

This lemma supplies the finite certificate families that make the structured cone theorem computational. It identifies the relevant extreme directions of each cone directly in terms of coordinates, prefix steps, suffix steps, and block steps.

Theorem 2 (Exact criteria for structured decrement cones). *For any agent i , robust-PS over the structured cones is characterised as follows.*

1. Over K^{all} , robust-PS holds if and only if

$$\Delta_i(k) \geq 0 \quad \forall k = 1, \dots, n-1.$$

2. Over K^{front} , robust-PS holds if and only if

$$\sum_{k=1}^m \Delta_i(k) \geq 0 \quad \forall m = 1, \dots, n-1.$$

3. Over K^{tail} , robust-PS holds if and only if

$$\sum_{k=m}^{n-1} \Delta_i(k) \geq 0 \quad \forall m = 1, \dots, n-1.$$

4. Over $K^{B\text{-front}}$, robust-PS holds if and only if

$$\sum_{j=1}^m S_j(\Delta_i) \geq 0 \quad \forall m = 1, \dots, J.$$

5. Over $K^{B\text{-tail}}$, robust-PS holds if and only if

$$\sum_{j=m}^J S_j(\Delta_i) \geq 0 \quad \forall m = 1, \dots, J.$$

For robust-RSD, reverse the displayed inequalities, or equivalently apply the same criteria to $-\Delta_i$.

This is the paper's main exact criterion theorem. It turns broad classes of economically interpretable utilities into concrete coordinate, prefix, suffix, and block-sum checks.

Proof sketch. Combine Abel representation with the generator lemma. Every admissible decrement vector in a structured cone is a nonnegative combination of step generators, so the welfare sign is determined by the corresponding certificate family. Necessity follows by selecting the single generator whose certificate is violated.

Corollary 2 (Finite-certificate sign-crossing criteria for $\mathcal{S}_D$). *For the structured cones in Theorem 2, the utility-sensitive region is characterised by mixed signs in the relevant finite certificate family.*

1. For K^{all} , $\Delta_i \in \mathcal{S}_D$ if and only if the coordinates $\Delta_i(k)$ include at least one positive and at least one negative value.
2. For K^{front} , $\Delta_i \in \mathcal{S}_D$ if and only if the prefix sums

$$\sum_{k=1}^m \Delta_i(k), \quad m = 1, \dots, n-1,$$

include at least one positive and at least one negative value.

3. For K^{tail} , $\Delta_i \in \mathcal{S}_D$ if and only if the suffix sums

$$\sum_{k=m}^{n-1} \Delta_i(k), \quad m = 1, \dots, n-1,$$

include at least one positive and at least one negative value.

4. For $K^{B\text{-front}}$ and $K^{B\text{-tail}}$, replace prefix and suffix sums by the corresponding block-prefix and block-suffix certificate values.

This corollary makes the utility-sensitive set concrete for the structured cones: indeterminacy is exactly sign crossing in the finite certificate family.

Corollary 3 (Full cone and dominance). *Robust dominance over K^{all} is exactly first-order stochastic dominance of the PS rank lottery over the RSD rank lottery.*

This corollary identifies the full cone with the familiar FOSD benchmark.

Corollary 4 (Non-nesting of front and tail cones). *If $n \geq 3$, then K^{front} and K^{tail} are not nested. Their intersection is the constant-decrement ray*

$$\{c(1, \dots, 1) : c \geq 0\}.$$

This clarifies that front and tail criteria are not merely stronger and weaker versions of the same restriction; they encode different institutional priorities.

Corollary 5 (Aggregate robust dominance). *Let*

$$\bar{\Delta} = \frac{1}{n} \sum_{i=1}^n \Delta_i.$$

For any closed convex cone $K \subseteq \mathbb{R}_+^{n-1}$, average welfare under PS is weakly higher than under RSD for every common decrement vector $d \in K$ if and only if

$$\bar{\Delta} \in K^*.$$

For the structured cones in Theorem 2, this is equivalent to applying the same coordinate, prefix, suffix, or block-sum checklist to $\bar{\Delta}$.

This is the aggregate version of the cone framework. It shows that the same duality and certificate logic survives when the welfare comparison is averaged across agents.

7 Simulation Methodology

The simulation has two distinct layers. The first layer implements the exact theory on synthetic finite markets. For each generated preference profile, the procedure is:

$$\text{profile} \rightarrow \text{PS and RSD rank lotteries} \rightarrow \Delta_i \rightarrow \text{cone classification.}$$

The generated grid uses market sizes

$$n \in \{3, 4, 5, 6, 8, 10, 15, 20, 30\}$$

and three preference environments: impartial culture, correlated preferences, and clustered preferences. For each profile and agent, the algorithm computes the cumulative gap vector Δ_i . It then applies the exact criteria in Theorem 2.

The classifications are:

- robust-PS: all relevant cone certificates are nonnegative and at least one is positive;
- robust-RSD: all relevant cone certificates are nonpositive and at least one is negative;
- equal: all relevant certificates are zero;
- utility-sensitive: the chosen envelope contains admissible directions favoring both mechanisms.

The RSD lottery is exact for small markets and Monte Carlo estimated for larger markets. Therefore the cone labels are exact conditional on the computed Δ_i ; for larger n , they inherit Monte Carlo uncertainty from the RSD estimates. No regression coefficient or causal test is used as a main claim. The numerical exercise is a prevalence map for the exact cone criteria.

The second layer is diagnostic. The simulation evaluates the stored power-family grid

$$\alpha \in \{0.5, 0.75, 1, 1.5, 2, 3\}.$$

Reported object	Input used	Theorem status	Interpretation
Cone labels	Δ_i , certificates	Exact given Δ_i	Robust dominance over an envelope
Cone slack	Δ_i , certificates	Exact given Δ_i	Distance from certificate boundary
Sensitivity geometry	Δ_i , certificates	Exact given Δ_i	Sign-crossing inside $\mathcal{S}_D$
Power-grid support	Stored EU gaps	Finite diagnostic	Benchmark-family direction
Environment shares	Synthetic labels, cone status	Descriptive	Prevalence map, not causal evidence

Table 2: Provenance of reported simulation objects. Exact labels are computed from Δ_i and the certificate criteria; power-grid support uses stored benchmark utility gaps only after the cone labels have been computed.

For each profile-agent observation, the analysis computes the fraction of these six benchmark utilities for which the stored welfare gap $EU_i^{\text{PS}} - EU_i^{\text{RSD}}$ is nonnegative. Equal weighting over this grid is only a benchmark summary; it is not a canonical prior over cardinal utilities.

Remark 1 (Power utilities as a benchmark slice). *The common power family of cardinal rank utilities is not itself a cone. For rank utilities of the form $w_\alpha(k) = -(k - 1)^\alpha$, the decrement vector is front-weighted for $0 < \alpha < 1$, constant at $\alpha = 1$, and tail-weighted for $\alpha > 1$. This makes the power family a useful pointwise slice through the admissible envelopes, but the cone criteria are stronger: they classify all utilities inside the chosen envelope, not only selected values of α .*

For a finite benchmark family D_0 , the same upper-contour representation also gives a secondary support score

$$\rho_{D_0}(\Delta_i) = \frac{1}{|D_0|} \sum_{d \in D_0} \mathbf{1}\{d^\top \Delta_i \geq 0\}.$$

This is not an ordinal-robustness criterion. It is a diagnostic for utility-sensitive cases: after the cone test fails to give unanimous agreement over an envelope, ρ_{D_0} summarises how the selected benchmark utilities rank PS against RSD. The simulation section uses this finite-family version only after the exact cone classifications have been computed.

Cone	Robust-PS	Robust-RSD	Utility-sensitive	Equal
Full positive cone	20.3%	3.5%	70.3%	5.9%
Front-weighted cone	57.6%	7.4%	29.1%	5.9%
Tail-weighted cone	64.3%	4.4%	25.4%	5.9%

Table 3: Overall cone classifications. Shares are rounded to one decimal percentage point.

8 Simulation Results

The simulation section asks two questions. First, how often does a computed Δ_i fall in a robust region rather than in the utility-sensitive set $\mathcal{S}_D$? Second, when $\Delta_i \in \mathcal{S}_D$, what kind of certificate disagreement produces the sensitivity?

Three patterns emerge. First, unrestricted first-order stochastic dominance is demanding: the full positive cone leaves most comparisons in $\mathcal{S}_D$. Second, economically structured utility classes resolve much of this indeterminacy: front- and tail-weighted envelopes classify many observations that the full cone leaves unsigned. Third, among observations that remain in $\mathcal{S}_D$, certificate signs are usually asymmetric rather than evenly split; many utility-sensitive cases contain many positive certificates and a smaller number of negative certificates.

Table 3 decomposes the comparison space into robust regions, the equal boundary, and $\mathcal{S}_D$ for each admissible envelope. The full cone is demanding: about 70.3 percent of observations are utility-sensitive under ordinary first-order dominance. The front and tail envelopes resolve much of that ambiguity. Under the front-weighted cone, 57.6 percent of observations are robust-PS, 7.4 percent are robust-RSD, and 29.1 percent remain in $\mathcal{S}_D$. Under the tail-weighted cone, 64.3 percent are robust-PS, 4.4 percent are robust-RSD, and 25.4 percent remain in $\mathcal{S}_D$.

Table 4 opens up the utility-sensitive region. The sign-crossing cases are not balanced knife-edge disagreements. For example, full-cone utility-sensitive observations average 14.3 positive certificates and 4.1 negative certificates; front- and tail-cone sensitive cases show a similar asymmetry. This is the empirical counterpart of Corollary 2: $\mathcal{S}_D$ is a structured region of conflicting certificate signs, not an unclassified residual.

Cone	Obs. in $\mathcal{S}_D$	$\mathcal{S}_D$ share	Mean + cert.	Mean – cert.	Mean spread	Median spread
Full positive cone	63,910	70.3%	14.3	4.1	0.065	0.058
Front-weighted cone	26,490	29.1%	15.7	3.2	0.253	0.215
Tail-weighted cone	23,054	25.4%	13.6	3.1	0.207	0.171

Table 4: Sign-crossing geometry inside $\mathcal{S}_D$. Positive and negative certificate counts are averaged over observations that are utility-sensitive for the relevant cone. Certificate spread is the difference between the largest and smallest certificate value.

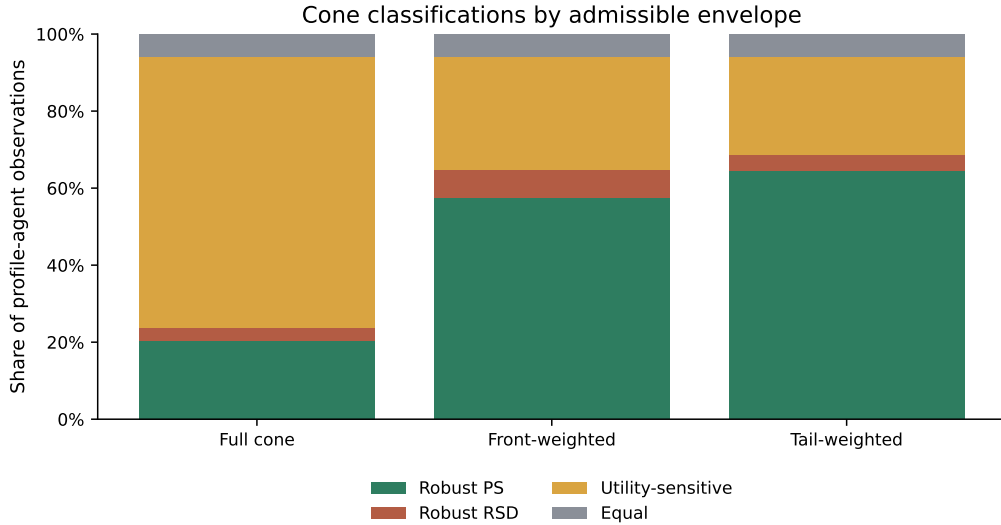


Figure 1: Overall cone classification shares. The utility-sensitive share is the empirical size of $\mathcal{S}_D$ for each admissible envelope. The full positive cone is the strict FOSD benchmark; front and tail envelopes classify many observations that are utility-sensitive under the full cone.

Figure 2 reports the same classification logic by market size for the uniformly random preference environment. It adds a finite-market view of the theorem’s implementation: as n changes, the share of observations that fall into robust-PS, robust-RSD, equal, and $\mathcal{S}_D$ changes differently across cones. The figure can be used a prevalence map for the simulated finite markets, not as an asymptotic convergence theorem.

The interpretation is not that PS universally dominates RSD. Rather, restricted utility envelopes make the welfare question more determinate. The front cone asks whether the PS-RSD cumulative gap is favorable for all utilities that put greater marginal emphasis on top-rank improvements. The tail cone asks the corresponding question for utilities that put greater marginal emphasis on avoiding low ranks. Because the cones are not nested, the

Cone Classifications as a Function of Market Size

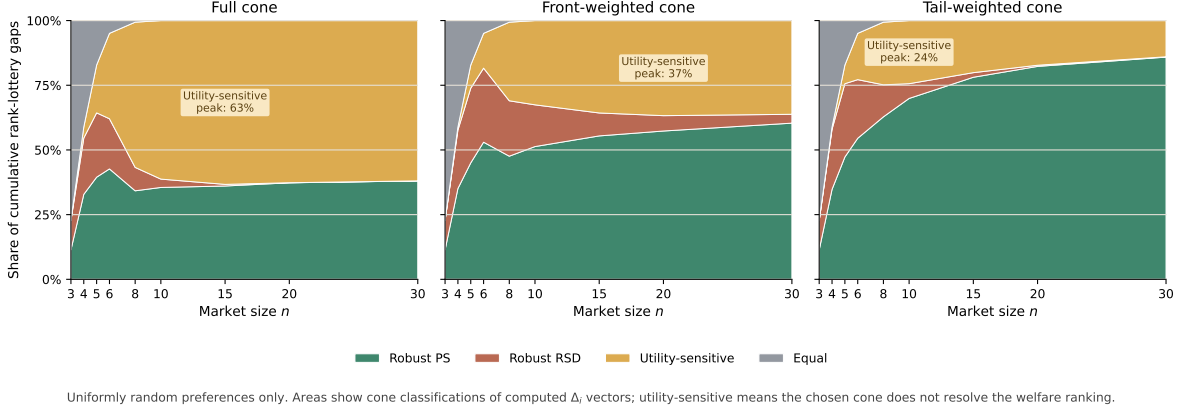


Figure 2: Resolution of welfare conflict as a function of market size. Areas show the share of computed cumulative rank-lottery gaps in the robust regions, the equal boundary, and the utility-sensitive set $\mathcal{S}_D$ for each cone under uniformly random preferences.

Cone	Environment	Robust-PS	Robust-RSD	Utility-sensitive	Equal
K^{front}	Clustered	53.1%	10.2%	31.6%	5.0%
K^{front}	Correlated	66.1%	0.7%	25.5%	7.7%
K^{front}	Impartial culture	53.5%	11.2%	30.3%	5.1%
K^{tail}	Clustered	65.3%	6.2%	23.5%	5.0%
K^{tail}	Correlated	55.2%	1.0%	36.2%	7.7%
K^{tail}	Impartial culture	72.6%	5.9%	16.4%	5.1%

Table 5: Front and tail cone classifications by generated environment.

shares are not a ranking of one cone as stronger than the other; they are classifications under different institutional priorities.

Table 5 shows that these robust regions differ by preference environment. Correlated markets have the highest robust-PS share under the front-weighted envelope, while impartial-culture markets have the highest robust-PS share under the tail-weighted envelope. This reflects a geometric property the cone criteria can detect. Therefore, choice of admissible envelope is sensitive to preference structure and not just some abstract mathematical concept.

Tiered cones provide the institutional generalisation behind Theorem 2. When cutoffs are grouped into broad tiers, the theorem requires block-prefix or block-suffix sums instead of rank-by-rank prefix or suffix sums. In the tercile specification, the tiered-front envelope

Sample	Obs.	Mean ρ	Median ρ	PS all six	RSD all six	Mixed
All observations	90,900	90.8%	100.0%	80.2%	5.7%	8.2%
Full-cone utility-sensitive	63,910	91.8%	100.0%	85.3%	3.1%	11.6%
K^{front} utility-sensitive	26,490	92.0%	100.0%	82.7%	2.6%	14.7%
K^{tail} utility-sensitive	23,054	82.5%	100.0%	71.7%	6.4%	21.8%

Table 6: Power-grid support diagnostic. Here ρ is the fraction of the six benchmark power utilities with $EU_i^{\text{PS}} - EU_i^{\text{RSD}} \geq 0$. The one-sided columns report strict finite-grid labels: all six benchmark utilities favor PS, or all six favor RSD. All-equal cases are counted in neither one-sided column; “Mixed” means at least one benchmark utility favors each mechanism.

classifies 81.4 percent of eligible observations as robust-PS, 8.9 percent as robust-RSD, 6.3 percent as utility-sensitive, and 3.4 percent as equal. The tiered-tail envelope yields 81.3 percent robust-PS, 5.1 percent robust-RSD, 10.2 percent utility-sensitive, and 3.4 percent equal. These figures are not a separate empirical claim; they illustrate the block-sum part of the same theorem.

8.1 Power-Grid Support inside $\mathcal{S}_D$

The cone labels identify where the welfare comparison is invariant over an entire admissible envelope. For observations that remain in $\mathcal{S}_D$, the power-family grid gives a secondary upper-contour diagnostic: it asks how many of the six benchmark cardinal utilities favor PS. Table 6 reports this diagnostic. The first row reports all profile-agent observations; the remaining rows restrict to observations that are utility-sensitive under each cone. These filtered rows are separate diagnostics and are not mutually exclusive.

The diagnostic should not be read as a replacement for robust dominance. It uses a finite benchmark grid rather than a full admissible cone or a canonical probability measure over utilities. Its role is to describe the remaining ambiguity after the exact cone criteria have been applied. In this sample, many observations in $\mathcal{S}_D$ are nevertheless one-sided along the power grid: among full-cone utility-sensitive observations, 85.3 percent have all six benchmark power utilities favoring PS. Tail utility-sensitive cases are less one-sided, with a higher mixed share of 21.8 percent. Thus the upper-contour diagnostic describes the direction of the remaining

benchmark-family ambiguity inside cases where the exact cone classification finds structured admissible disagreement.

9 Conclusion

This thesis shows how welfare comparisons between PS and RSD can be made without choosing a single cardinal representation, but rather using features the family of cardinal representations may have. It follows on that the main contribution is a restricted dominance framework for finite assignment markets. The framework identifies whether PS or RSD robustly dominates the other or whether the comparison is utility sensitive. The latter case isn't a failure of the framework, rather it is a way to tell when welfare ranking depends on substantive assumptions about how agents value improvement at and between different ranks. This follows onto the economically meaningful utility restrictions with front and tail weighted and tiered decrement classes translate into prefix and suffix sums. These criteria allow policymakers to make agents and institutional priorities explicit. The simulations illustrate how the theory operates for a synthetic finite market. Whilst it does not constitute empirical validation, it shows the robust dominance and utility sensitivity distinction in a quantitatively meaningful way. This thesis has several limitations. First, it is a truthful-report welfare benchmark, so we make the assumption agents report honestly since PS is not strategy proof but RSD. Second, the admissible decrement classes are imposed by the analyst: front-weighted, tail-weighted, and tiered cones are economically interpretable. The choice of this would likely be justified by normative priorities defined by institution or the analyst themselves. Finally, for larger markets the RSD lottery is Monte Carlo estimated, so classifications close to certificate boundaries may inherit simulation error.

There are several natural extensions to this thesis. The same restricted dominance perspective can be applied to other random assignment mechanisms or environments in which admissible restrictions differ across agents or institutions. It could be connected to an

empirical study. During the course of this paper, a block additivity lemma was also produced in the appendix, which could be expanded upon. This looks at finite submarkets within larger markets and the implications for asymptotic equivalence.

A Proofs

A.1 Proof of Proposition 1

Proof. Let $a_k = p_i^{\text{PS}}(k) - p_i^{\text{RSD}}(k)$, and let

$$A_k = \sum_{r=1}^k a_r.$$

Then $A_k = \Delta_i(k)$ for $k < n$, and $A_n = 0$ because both mechanisms assign one object with probability one. Summation by parts gives

$$\sum_{k=1}^n a_k w(k) = A_n w(n) - \sum_{k=1}^{n-1} A_k (w(k+1) - w(k)).$$

Since $A_n = 0$, this equals

$$\sum_{k=1}^{n-1} \Delta_i(k) (w(k) - w(k+1)) = d_w^\top \Delta_i.$$

□

A.2 Proof of Theorem 1

Proof. The condition $\phi_D(\Delta_i) \geq 0$ is equivalent to $d^\top \Delta_i \geq 0$ for every $d \in D$. Similarly, $\psi_D(\Delta_i) \leq 0$ is equivalent to $d^\top \Delta_i \leq 0$ for every $d \in D$. If the lower envelope is negative and the upper envelope is positive, the sign of $d^\top \Delta_i$ varies over admissible decrement vectors. If both envelopes equal zero, every admissible welfare gap equals zero. □

A.3 Proof of Corollary 1

Proof. The claim follows by weakening the quantifier from all $d \in K$ to all $d \in D$. $\square$

A.4 Proof of Proposition 2

Proof. Because $D_0 \subseteq D_1$, taking the infimum over the larger family can only weakly decrease the value, while taking the supremum over the larger family can only weakly increase it. This proves the first display. If $\phi_{D_1}(\Delta) \geq 0$, then $\phi_{D_0}(\Delta) \geq \phi_{D_1}(\Delta) \geq 0$, which gives the robust-PS inclusion. The robust-RSD inclusion is identical with ψ . Finally, if $\phi_{D_0}(\Delta) < 0 < \psi_{D_0}(\Delta)$, then $\phi_{D_1}(\Delta) \leq \phi_{D_0}(\Delta) < 0$ and $\psi_{D_1}(\Delta) \geq \psi_{D_0}(\Delta) > 0$, so Δ also belongs to the larger utility-sensitive set. $\square$

A.5 Proof of Proposition 3

Proof. By definition, $\Delta_i \in K^*$ exactly when $d^\top \Delta_i \geq 0$ for all $d \in K$. This is robust-PS. Replacing Δ_i by $-\Delta_i$ gives the robust-RSD statement. $\square$

A.6 Proof of Proposition 4

Proof. If the generator inequalities hold and $d \in K$, then

$$d = \sum_{j=1}^m \lambda_j g^j \quad \text{for some } \lambda_j \geq 0.$$

Therefore

$$d^\top \Delta_i = \sum_{j=1}^m \lambda_j (g^j)^\top \Delta_i \geq 0.$$

This proves robust-PS. The converse follows because each generator g^j belongs to K . The robust-RSD statement is identical with all inequalities reversed. $\square$

A.7 Proof of Proposition 5

Proof. If $\Delta \in \mathcal{S}_D$, then $\phi_D(\Delta) < 0$, so not every $d^\top \Delta$ can be nonnegative. Hence there exists $d \in D$ with $d^\top \Delta < 0$. Similarly, $\psi_D(\Delta) > 0$ implies that there exists $d' \in D$ with $(d')^\top \Delta > 0$. The converse is immediate: the existence of a negative admissible welfare gap implies $\phi_D(\Delta) < 0$, and the existence of a positive admissible welfare gap implies $\psi_D(\Delta) > 0$. The half-space identities are the same statement written as intersections and unions of the sign regions generated by the linear welfare functionals. $\square$

A.8 Proof of Lemma 1

Proof. The full-cone statement is immediate from the coordinate basis.

For K^{front} , take $d \in K^{\text{front}}$, set $d(n) = 0$, and define

$$\lambda_m = d(m) - d(m+1) \geq 0.$$

Then

$$d(k) = \sum_{m=k}^{n-1} \lambda_m = \sum_{m=1}^{n-1} \lambda_m q^m(k).$$

Thus d is a nonnegative combination of the prefix-step vectors q^m . The reverse inclusion is immediate because each q^m is nonnegative and weakly decreasing.

For K^{tail} , set $d(0) = 0$ and define

$$\mu_m = d(m) - d(m-1) \geq 0.$$

Then

$$d(k) = \sum_{m=1}^k \mu_m = \sum_{m=1}^{n-1} \mu_m r^m(k).$$

The reverse inclusion follows because each r^m is nonnegative and weakly increasing.

For $K^{B\text{-front}}$, write $d(k) = c_j$ on block B_j , set $c_{J+1} = 0$, and define $\lambda_m = c_m - c_{m+1} \geq 0$.

Then

$$c_j = \sum_{m=j}^J \lambda_m,$$

so $d = \sum_{m=1}^J \lambda_m Q^m$. The reverse inclusion holds because each Q^m is block-constant and weakly decreasing across blocks. The proof for $K^{B\text{-tail}}$ is identical with $c_0 = 0$, $\mu_m = c_m - c_{m-1} \geq 0$, and $d = \sum_{m=1}^J \mu_m R^m$. $\square$

A.9 Proof of Theorem 2

Proof. The full cone is immediate. Since $K^{\text{all}} = \mathbb{R}_+^{n-1}$, robust-PS over K^{all} requires $d^\top \Delta_i \geq 0$ for every nonnegative d . This holds if and only if each coordinate $\Delta_i(k)$ is nonnegative. If some coordinate is negative, choosing $d = e^k$ violates robust-PS.

For the front-weighted cone, take any $d \in K^{\text{front}}$. Because d is weakly decreasing and nonnegative, set $d(n) = 0$ and define

$$\lambda_m = d(m) - d(m+1), \quad m = 1, \dots, n-1.$$

Each λ_m is nonnegative. For every cutoff k ,

$$d(k) = \lambda_k + \lambda_{k+1} + \dots + \lambda_{n-1} = \sum_{m=k}^{n-1} \lambda_m.$$

Substituting this step decomposition into the Abel representation gives

$$d^\top \Delta_i = \sum_{k=1}^{n-1} \Delta_i(k) \sum_{m=k}^{n-1} \lambda_m.$$

Interchanging the order of summation yields

$$d^\top \Delta_i = \sum_{m=1}^{n-1} \lambda_m \left(\sum_{k=1}^m \Delta_i(k) \right).$$

If every prefix sum is nonnegative, this expression is nonnegative for every choice of nonnegative

coefficients λ_m , and hence for every $d \in K^{\text{front}}$. Conversely, if

$$\sum_{k=1}^m \Delta_i(k) < 0$$

for some m , choose $\lambda_m > 0$ and set all other λ 's equal to zero. The corresponding step decrement vector is in K^{front} , and it makes $d^\top \Delta_i < 0$. Therefore robust-PS fails.

For the tail-weighted cone, take any $d \in K^{\text{tail}}$. Since d is weakly increasing and nonnegative, set $d(0) = 0$ and define

$$\mu_m = d(m) - d(m-1), \quad m = 1, \dots, n-1.$$

Each $\mu_m \geq 0$, and for every cutoff k ,

$$d(k) = \sum_{m=1}^k \mu_m.$$

The Abel representation becomes

$$d^\top \Delta_i = \sum_{k=1}^{n-1} \Delta_i(k) \sum_{m=1}^k \mu_m.$$

Interchanging the order of summation gives

$$d^\top \Delta_i = \sum_{m=1}^{n-1} \mu_m \left(\sum_{k=m}^{n-1} \Delta_i(k) \right).$$

If every suffix sum is nonnegative, the expression is nonnegative for every nonnegative vector μ , and hence for every $d \in K^{\text{tail}}$. If some suffix sum is negative, place positive mass on the corresponding μ_m and zero mass elsewhere. The resulting tail-weighted decrement vector makes $d^\top \Delta_i < 0$.

For the tiered-front cone, write $d(k) = c_j$ for $k \in B_j$. Set $c_{J+1} = 0$, and define

$$\lambda_m = c_m - c_{m+1} \geq 0, \quad m = 1, \dots, J.$$

Then

$$c_j = \sum_{m=j}^J \lambda_m.$$

The welfare gap is

$$\sum_{k=1}^{n-1} d(k) \Delta_i(k) = \sum_{j=1}^J c_j S_j(\Delta_i).$$

Substituting the step decomposition gives

$$\sum_{j=1}^J c_j S_j(\Delta_i) = \sum_{j=1}^J S_j(\Delta_i) \sum_{m=j}^J \lambda_m = \sum_{m=1}^J \lambda_m \sum_{r=1}^m S_r(\Delta_i).$$

Since the λ_m 's can be any nonnegative coefficients, the expression is nonnegative for all tiered-front decrement vectors if and only if every block-prefix sum is nonnegative. If one block-prefix sum is negative, isolate the corresponding λ_m to obtain a tiered-front decrement vector with negative welfare gap.

For the tiered-tail cone, set $c_0 = 0$ and define

$$\mu_m = c_m - c_{m-1} \geq 0, \quad m = 1, \dots, J.$$

Then $c_j = \sum_{m=1}^j \mu_m$, and

$$\sum_{j=1}^J c_j S_j(\Delta_i) = \sum_{m=1}^J \mu_m \sum_{r=m}^J S_r(\Delta_i).$$

This is nonnegative for every nonnegative μ_m if and only if every block-suffix sum is nonnegative. If one block-suffix sum is negative, isolate the corresponding μ_m to obtain a tiered-tail decrement vector with negative welfare gap.

The robust-RSD criteria follow by applying the same arguments to $-\Delta_i$. $\square$

A.10 Proof of Corollary 2

Proof. By Proposition 4, each structured cone is tested by a finite certificate family. If all certificate values are nonnegative, the comparison is robust-PS. If all are nonpositive, it is robust-RSD. If all are zero, the mechanisms are equal over the cone. The only remaining case is that at least one certificate value is positive and at least one is negative, which is exactly $\Delta_i \in \mathcal{S}_D$ by Proposition 5. $\square$

A.11 Proof of Corollary 3

Proof. First-order stochastic dominance over ranks requires $F_i^{\text{PS}}(k) \geq F_i^{\text{RSD}}(k)$ for every cutoff $k < n$. This is exactly $\Delta_i(k) \geq 0$ for every k , which is the K^{all} criterion in Theorem 2. $\square$

A.12 Proof of Corollary 4

Proof. For any n , a vector in both cones must be weakly decreasing and weakly increasing, hence constant. Conversely, every nonnegative constant vector is in both cones. To see non-nesting when $n \geq 3$, note that there are at least two decrement coordinates. Then $d = (1, 0, \dots, 0)$ is in K^{front} but not K^{tail} , while $d = (0, \dots, 0, 1)$ is in K^{tail} but not K^{front} . $\square$

A.13 Proof of Corollary 5

Proof. Average welfare gap equals

$$\frac{1}{n} \sum_i d^\top \Delta_i = d^\top \bar{\Delta}.$$

The result follows from Proposition 3 and Theorem 2. $\square$

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B AI Usage and Transparency Note

Gemini Pro 3.1 was used to format handwritten ideas into Latex where relevant. Codex (GPT-5.4) was used as a CLI assistant when coding i.e. market generation, producing

visualisations. Claude (Sonnet 4.6 - Free) was used to review the paper and logic i.e. give feedback and point out spelling errors.

Proofs and mathematics received aid from a friend and graduate student in Operations Research (MEng) prior to any AI usage. This involved checking my proofs and finding fallacies and some mathematical guidance. Propositions etc. are still my own.

C Interesting Findings

Lemma 2 (Exact block additivity). *Let the market be partitioned into agent-object blocks $\mathcal{B} = \{(A_r, O_r)\}_{r=1}^R$ with $|A_r| = |O_r|$. Suppose the partition is fully preference-separable: for every $i \in A_r$, every $o \in O_r$, and every $o' \notin O_r$,*

$$o \succ_i o'.$$

Let M_r denote the assignment submarket induced by block r . Then, after ordering agents and objects by block,

$$X^{\text{PS}}(M) = \bigoplus_{r=1}^R X^{\text{PS}}(M_r), \quad X^{\text{RSD}}(M) = \bigoplus_{r=1}^R X^{\text{RSD}}(M_r).$$

Consequently, for every agent $i \in A_r$, the PS-RSD cumulative gap vector computed in the full market is the same as the cumulative gap vector computed in the isolated block, after padding with zeros on cutoffs that only separate objects below the agent's own block.

This implies that the asymptotic convergence requires number of blocks to tend to infinity rather than just n per se.

Proof. Under full separability, no agent is ever assigned outside her own block.

For PS, suppose otherwise and let $t < 1$ be the first time any agent consumes outside her block. If this agent is in A_r , then before t the agents in A_r have consumed only from O_r , with total mass $|A_r|t$. Since $|A_r| = |O_r|$ and $t < 1$, some object in O_r remains. Full separability

therefore makes outside consumption impossible, a contradiction.

For RSD, the restriction of a uniform global priority order to A_r is uniform on A_r . When an agent in A_r chooses, fewer than $|A_r|$ agents from A_r have chosen before her, so some object in O_r remains; by separability she chooses in O_r . Thus global RSD decomposes into independent local RSDs.

Hence both mechanisms are block diagonal:

$$X^{\text{PS}}(M) = \bigoplus_r X^{\text{PS}}(M_r), \quad X^{\text{RSD}}(M) = \bigoplus_r X^{\text{RSD}}(M_r).$$

For $i \in A_r$, the full-market rank lottery agrees with the isolated-block rank lottery on O_r , and both mechanisms assign zero probability to lower outside-block ranks. The cumulative PS – RSD gap is therefore the isolated block gap, padded by zeros. $\square$